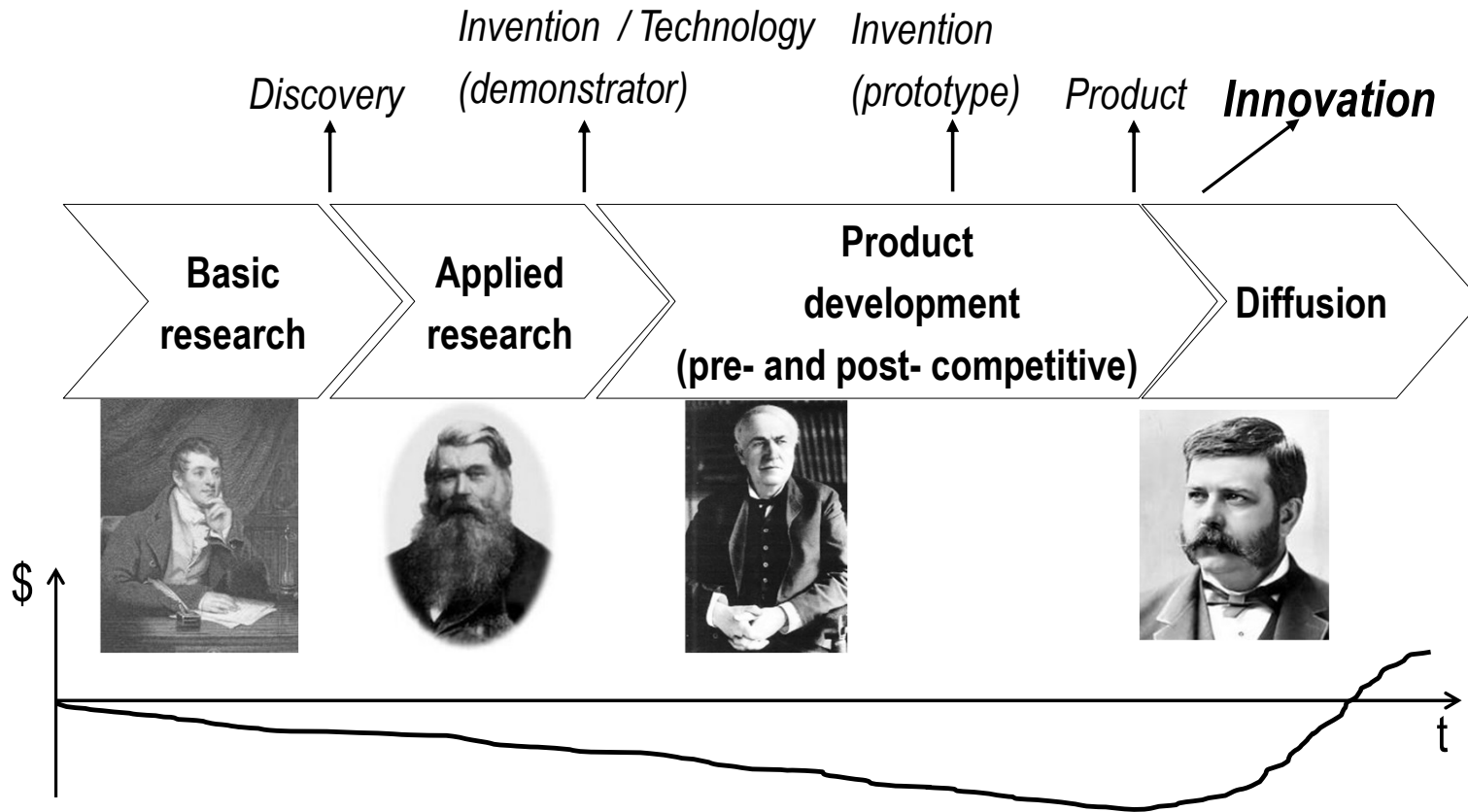


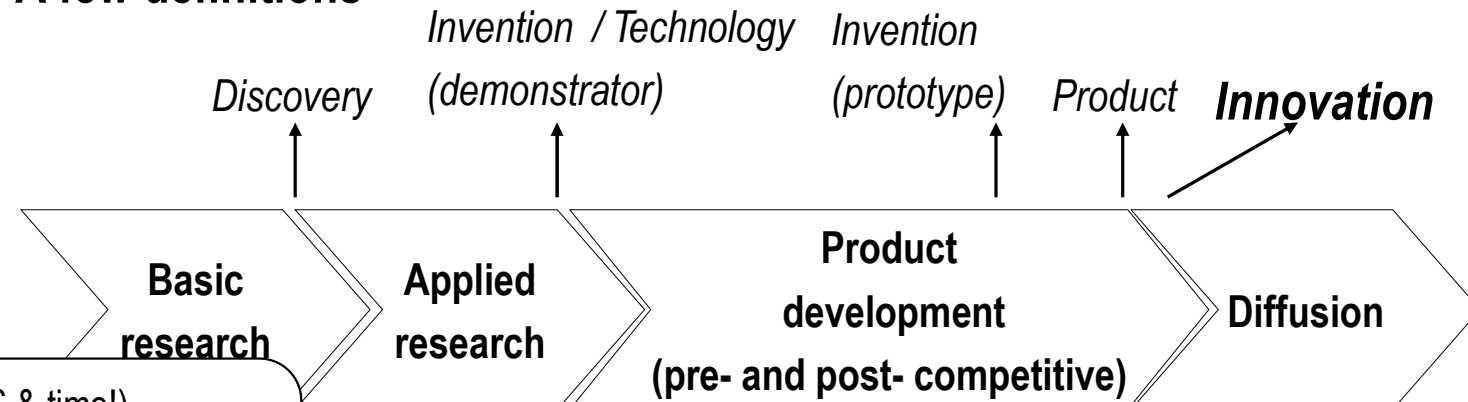
INVENTION AND INNOVATION

A few definitions



- Should you believe this "linear model"?
- Does "technology follow science" and "science precede technology"?
- Is it a 1:1 relationship, or is it m:n?

A few definitions



- Not enough ROI (\$ & time!)
- Keeping results secret would be socially suboptimal
- Markets for technology allow waiting

Risk & failure rate

Role of business

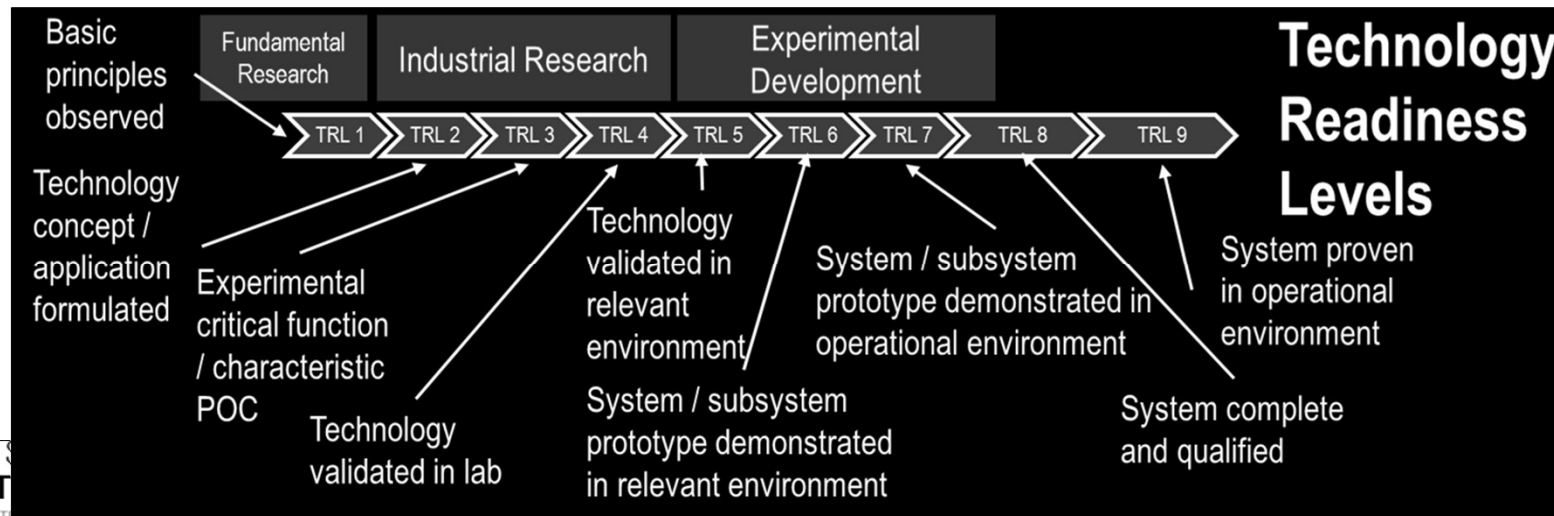
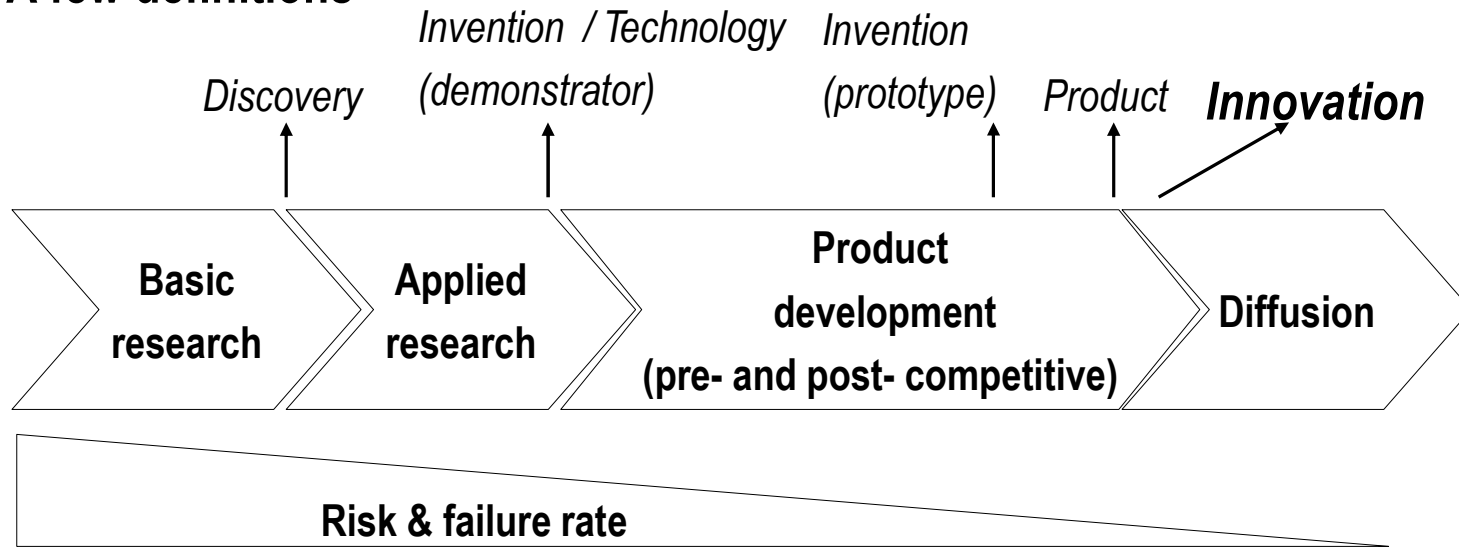
Role of academia and government

- funding
- public knowledge
- skilled workforce

Indirect actions can be effective

- Schumpeterian innovator-entrepreneurs (*widening* innovations)
- Schumpeter's large firms (*deepening* innovations)
- Chesbrough's Open Innovation networks and ecosystems ... and...
- Dougherty and Dunne's ecologies (*complex* innovations)
- Ramaswamy's co-creative customers (?)

A few definitions



Public funding

Policy	Rationale	Tools	Pros	Cons
Supply side / Direct funding	State provides resources that Market does not supply	Grants, low interest loans (e.g., Smart&Start, H2020 SME instrument)	Speed (depending on procedures) Ex-ante definition of budget	Allocational efficiency (picking winners based on proposals?) Bureaucracy (time and cost)
Supply side / Input Incentives	State increases ROI to Market by reducing immediate outflows or risk	Matching funds Tax breaks (e.g., for startup financing) Guarantees (e.g. for bank loans)	Speed Market involvement	Needs a market capable of selecting Risk of opportunistic behavior Cost of policy difficult to quantify ex-ante
Supply side / Output incentives	State increases ROI by boosting potential inflows, if and when results are achieved	Rebates to income tax (e.g., 10-year tax waivers on corporate income tax, tax exemptions for capital gains, "patent boxes")	Market involvement Lower risk for state (resources go to those who 'risk <i>and reap</i> ' if and only if, and after, benefits have accrued)	Needs a market capable of selecting Difficult to quantify ex-ante cost of policy (still a problem?)
Demand side / procurement	State stimulates innovation by creating an early market for innovation	Public Technology Procurement (e.g., US SBIR, DoD)	Award given to effective results and not to promises Creates a market and supply chain State benefits from improved technology	Requires an entrepreneurial culture and appropriate rules within civil service
Demand side / regulation, standardization	State creates and shapes the demand for innovation	Product-specific regulation (e.g., biodegradable bags, CO2 emissions for vehicles)	No immediate cost	Technical choices / targets can be non-neutral and difficult to achieve Risk of technological lock-in Requires a credible State and a capable Market

Public funding

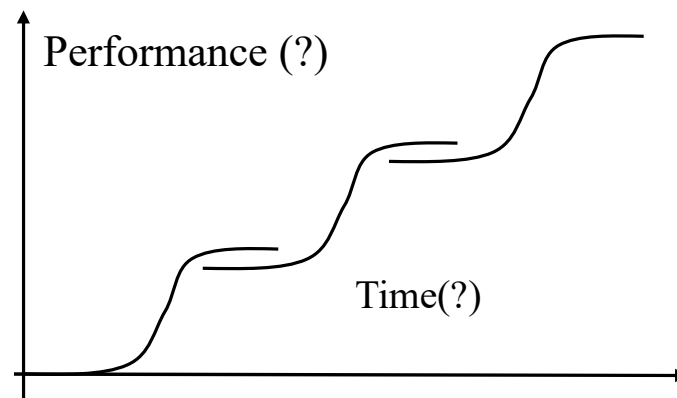
Policy	Rationale	Tools	Pros	Cons
Demand side/incentives	State reduces the cost/boosts the benefits of <i>adopting</i> the innovation	Direct incentives to adopters when purchasing (e.g., purchase rebates for Evs, hyperamortization for capital goods) Incentives to the usage (e.g., feed-in tariffs for renewable energy producers)	Strong and immediate impulse to adoption Award given to effective results and not to promises Creates a market and supply chain Society benefits from rapid diffusion	Applicable to technology that is market-ready Risk of benefiting foreign producers Cost can be high and difficult to quantify Risk of technological lock-in Induces volatility in demand before and after the policy is active Could be fiscally regressive, not generate additionality and lead to unused goods
Demand side / complementary assets	State facilitates and shapes the demand for innovation	Investment in complementary assets (e.g., BEV recharging stations)	High probability of having impact	No technological neutrality and risk of lock-in High cost
Non-specific	State improves general "business friendliness"	General reforms (tax, education, infrastructure, judicial system, education, etc.)	Stable and visible impact throughout the economy No ex ante directional choice w.r.t. industry or field	Time required

Private funding

Private financing	How does it work?	Pros	Cons
Bootstrapping	Use margins from operations to finance innovation	No dilution and total independence	Time required, entity and variability of margins
Debt	Get loans from bank or issue bonds... and then pay it back (or convert to equity, if the loan is convertible)	No dilution	Not acceptable for high-risk projects
Customer financing	Early customers finance your innovation (from pilots to reward crowdfunding)	Deep engagement with market needs No dilution Risk sharing	Narrow engagement with customer needs Can you find such customer(s)?
Business angels	Wealthy people with industry experience, alone or in club deals, provide "smart money" as equity (or future equity if SAFE is used) to high-risk startups	High risk acceptance Competencies and relations	Dilution Possible intrusiveness
Venture Capital	VC funds (GPs) raise money from investors (LPs) and provide staged financing to high-risk startups, supporting their growth and looking for an exit (IPO/trade sale). A few "home runs" compensate for failures and write-offs	High risk acceptance Need an exit	Dilution Need an exit
Private Equity	Like VC, but for less risky firms, might not need an exit and entails greater PE involvement in management	Risk acceptance	Dilution Loss of control
IPO	Raise money from public markets	Amount that can be raised Founders can keep control Liquidity of shares	Dilution Costly and complex process

Paradigms

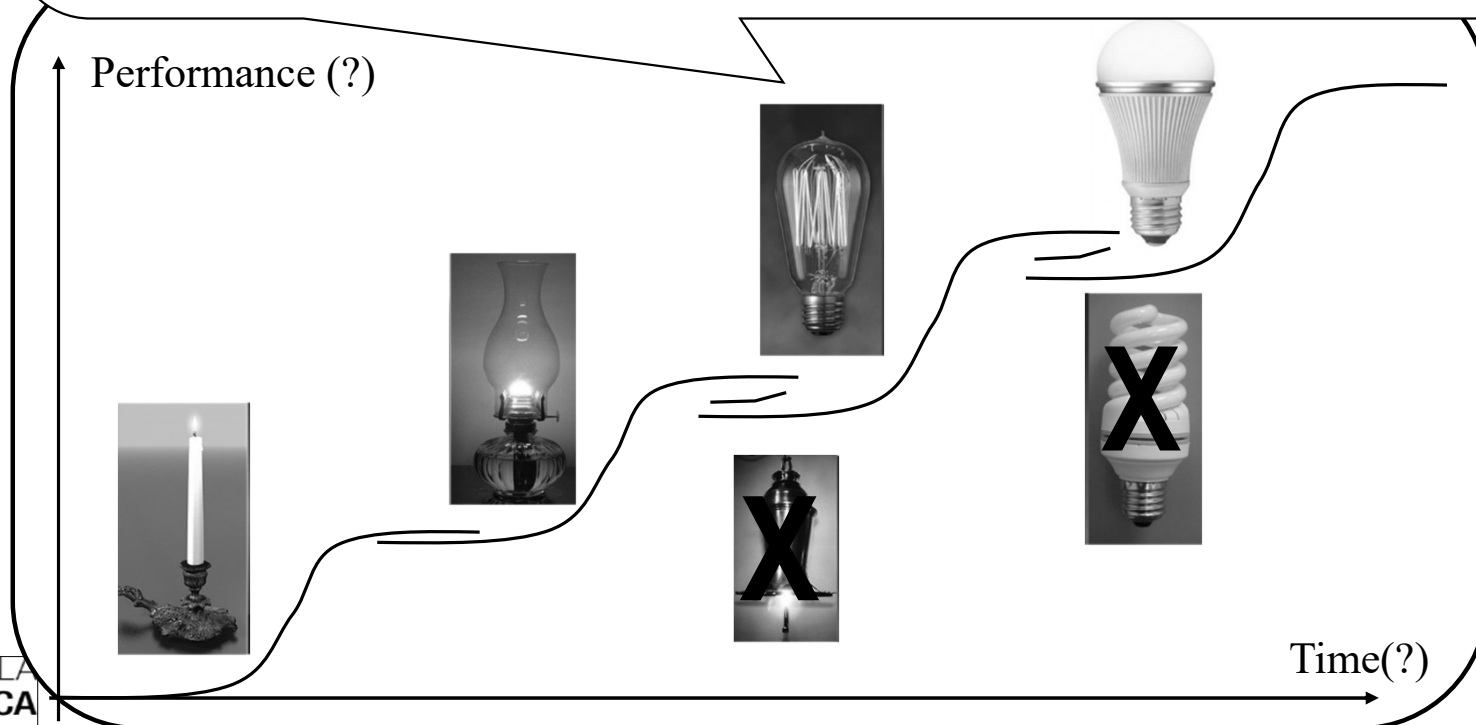
Does it look like a story characterized by continuous progress?



Paradigms

Not really! Dosi and Freeman show that Innovation follows a continuous alternation of

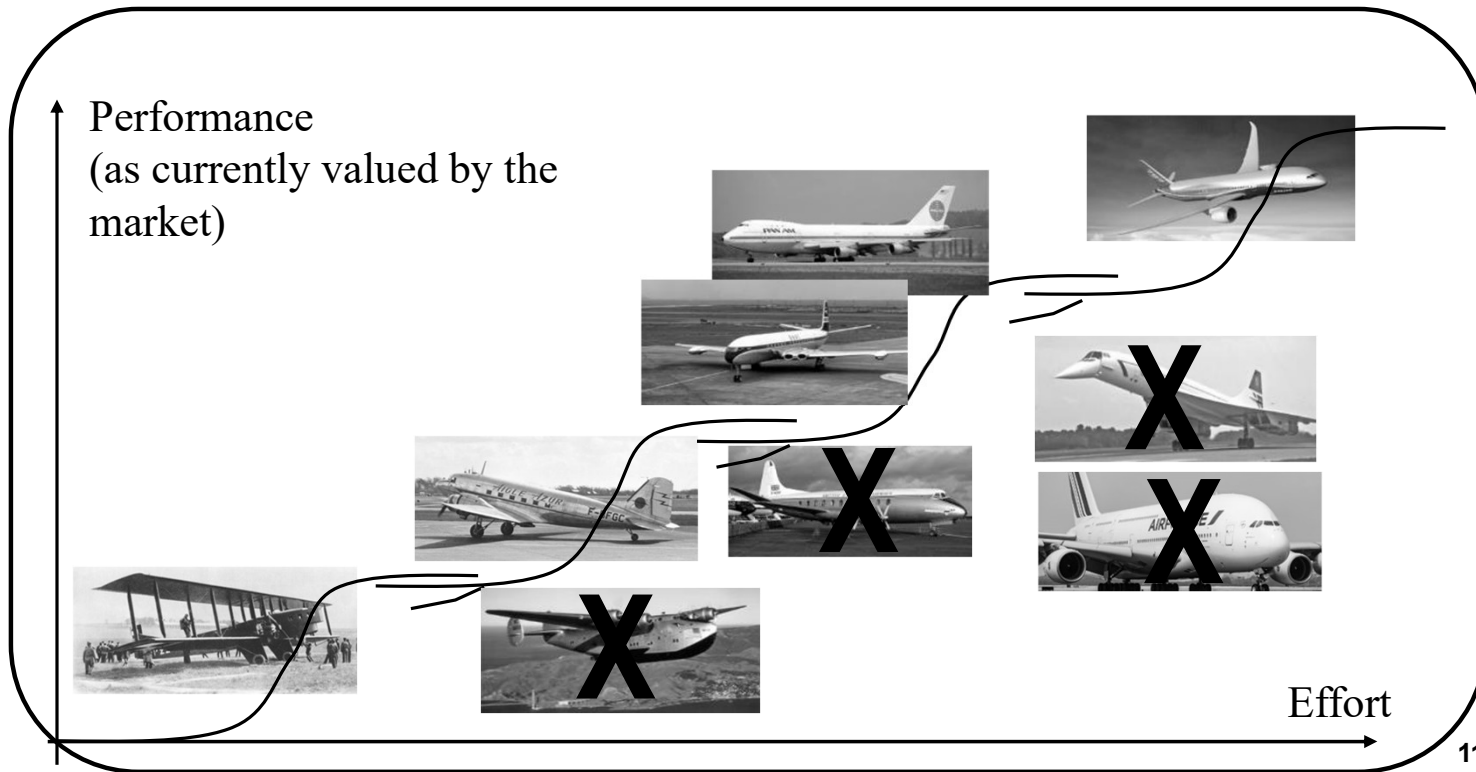
- Evolutionary and revolutionary change
- Performance s-curves
 - Bounded by technological limits
 - Initiated by tapping into new technical solutions (*technology push*)
 - Emerging after a period of uncertainty (revolution)
 - Evolving in accordance to the emergent paradigm (*demand pull*)



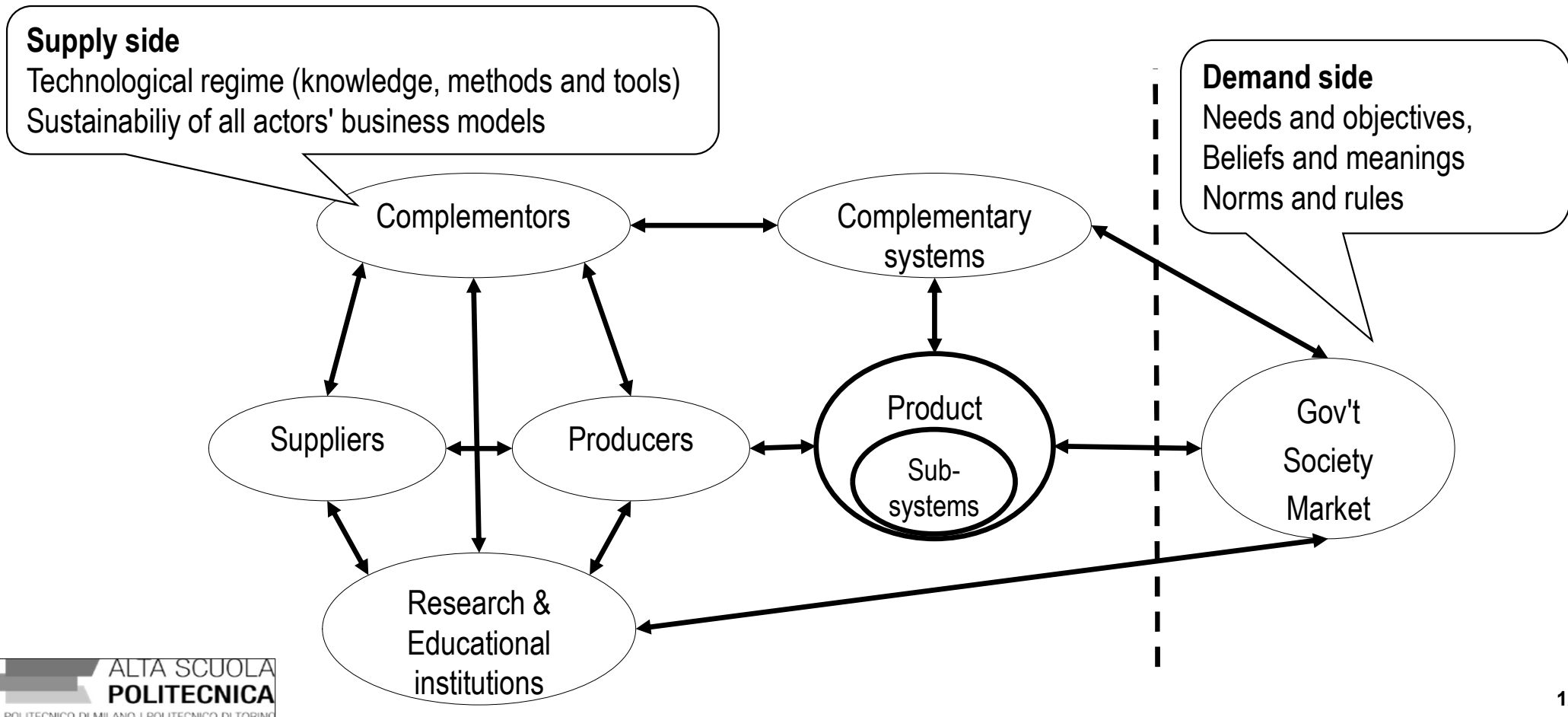


Airliner	Year	Airframe	Engines	Cruise speed (km/h)	Range (km)	Max pass.	Fuel eff. (km seat/l)
Flyer	1903	Wood + fabric	1 piston + propeller	48	-	1	-
Farman Goliath	1919	Wood + fabric	2 piston + propeller	120	400	14	13.4
Douglas DC-3	1936	Metal	2 piston + propeller	333	2400	32	8.4
Boeing 314	1938	Metal	4 piston + propeller	340	5900	74	23.5
Vickers Viscount 700	1948	Metal	4 turboprop	496	2220	48	15.3
De Havilland Comet	1949	Metal	4 turbojet	740	2400	44	3.56
Boeing 747-100, 200	1970	Metal	4 turbojet	893	12690	550	26.9
Concorde	1976	Metal	4 turbojet	2158	7222	120	6.4
Airbus 380	2007	Metal	4 turbofan	903	15460	853	31.6
Boeing 787	2011	Composites	2 turbofan	913	15400	440	36.6

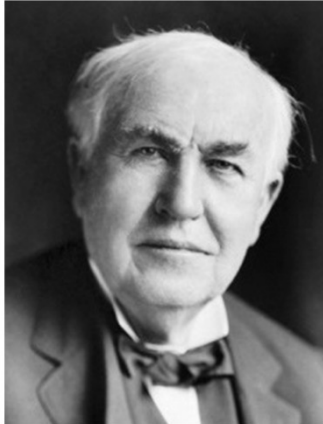
Paradigms



Paradigms



Paradigms



What were the competing
paradigms they proposed?
Which one won?
Why?

Paradigms

PEST analysis can help structure a sound understanding of what's going on

Political	Economic
Societal	Technological

Types of innovation

Criterion (look at...)	An "evolutionary" innovation is called	A "revolutionary" innovation is called
Product and the technical tradeoffs that define it	Incremental innovation	Radical innovation
Producer and its organization	Competence enhancing innovation	Competence destroying innovation
Product architecture	Peripheral innovation	Core innovation
Business impact	Sustaining innovation	Disruptive innovation



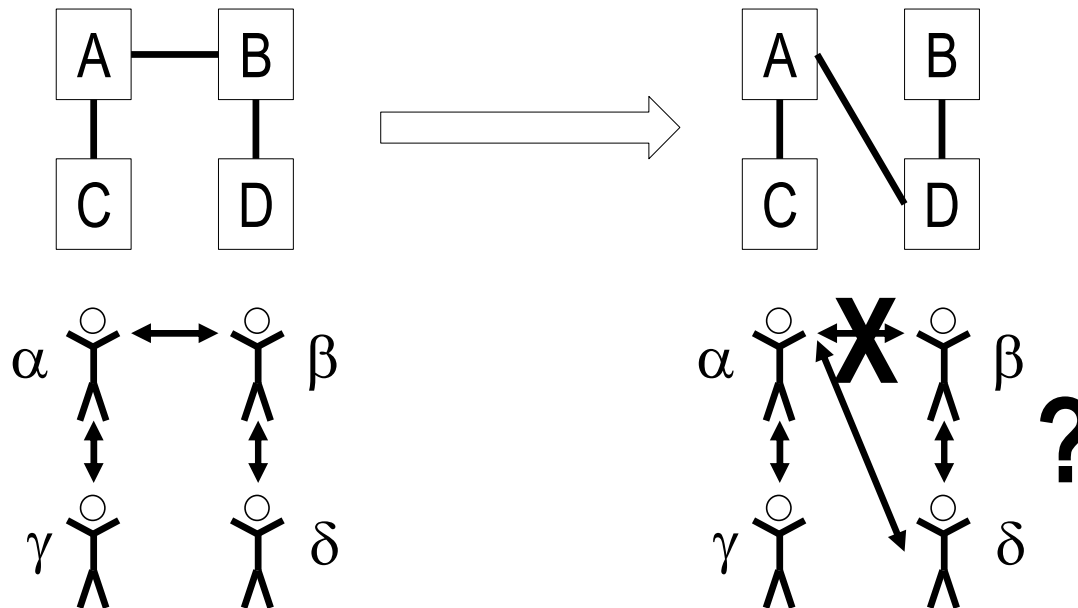
What kind of innovation is it?
It depends on who you are!



Types of innovation

Product architecture $\approx$ components
and their mutual relationships

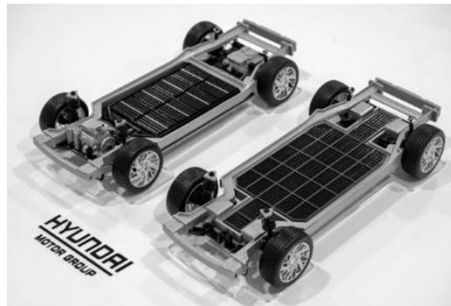
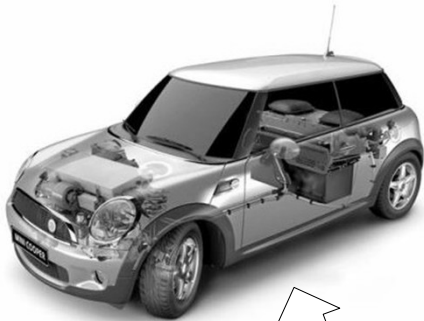
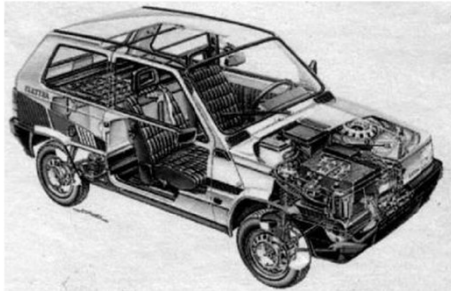
- Innovation and product architecture (Henderson and Clark)
 - Product architecture and the organization are coupled
 - What must happen if product architecture changes?



Types of innovation

- Innovation and product architecture (Henderson and Clark)
 - Product architecture and the organization are coupled
 - It is generally harder to change architecture than technology
 - Innovations often are not incremental or modular (but incumbents tend to downplay them as such)

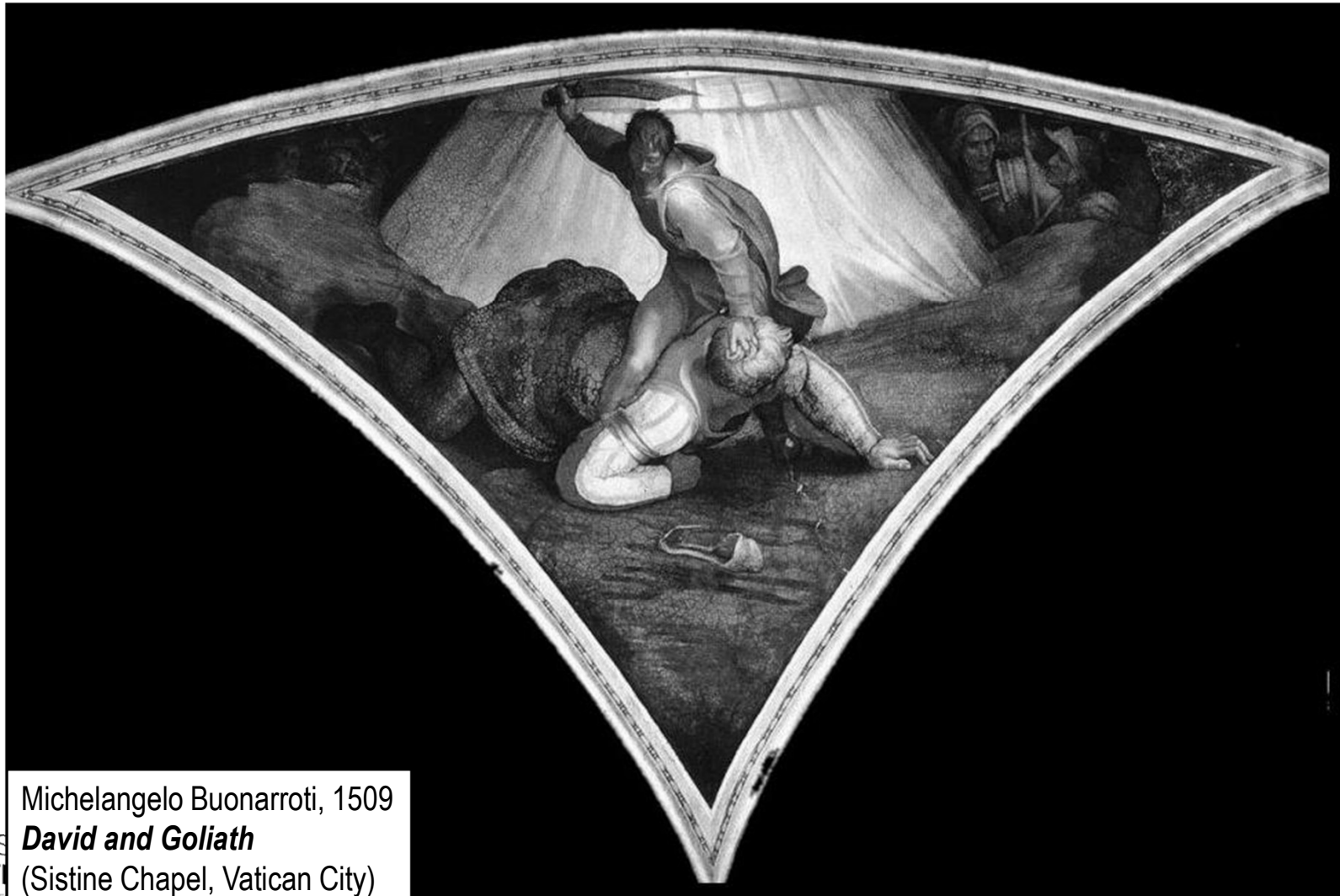
Relationships between components Reference technologies	Do not change	Change
Change	Modular innovation	Radical innovation
Do not change	Incremental innovation	Architectural innovation



Relationships between components Reference technologies	Do not change	Change
Change	Modular innovation	Radical innovation
Do not change	Incremental innovation	Architectural innovation



Types of innovation



Michelangelo Buonarroti, 1509
David and Goliath
(Sistine Chapel, Vatican City)

Types of innovation

- Radical innovations sometimes are disruptive and sometimes not
- How can innovations be *disruptive* (i.e. why do *incumbent* firms fail to face them)?

There are growing or new customer needs

AND

The old technology does not keep pace with them

AND

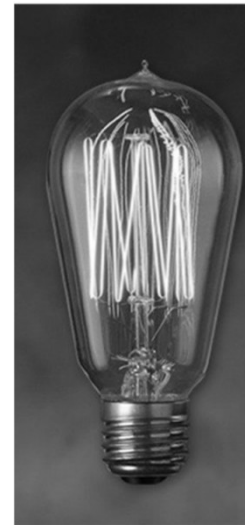
The new technology does

AND

Incumbents cannot effectively pursue the new technology because of misaligned resources and organizational inertia (*competency traps*)

Types of innovation

- Radical innovations sometimes are disruptive and sometimes not
- How can innovations be *disruptive* (i.e. why do *incumbent* firms fail to face them)?

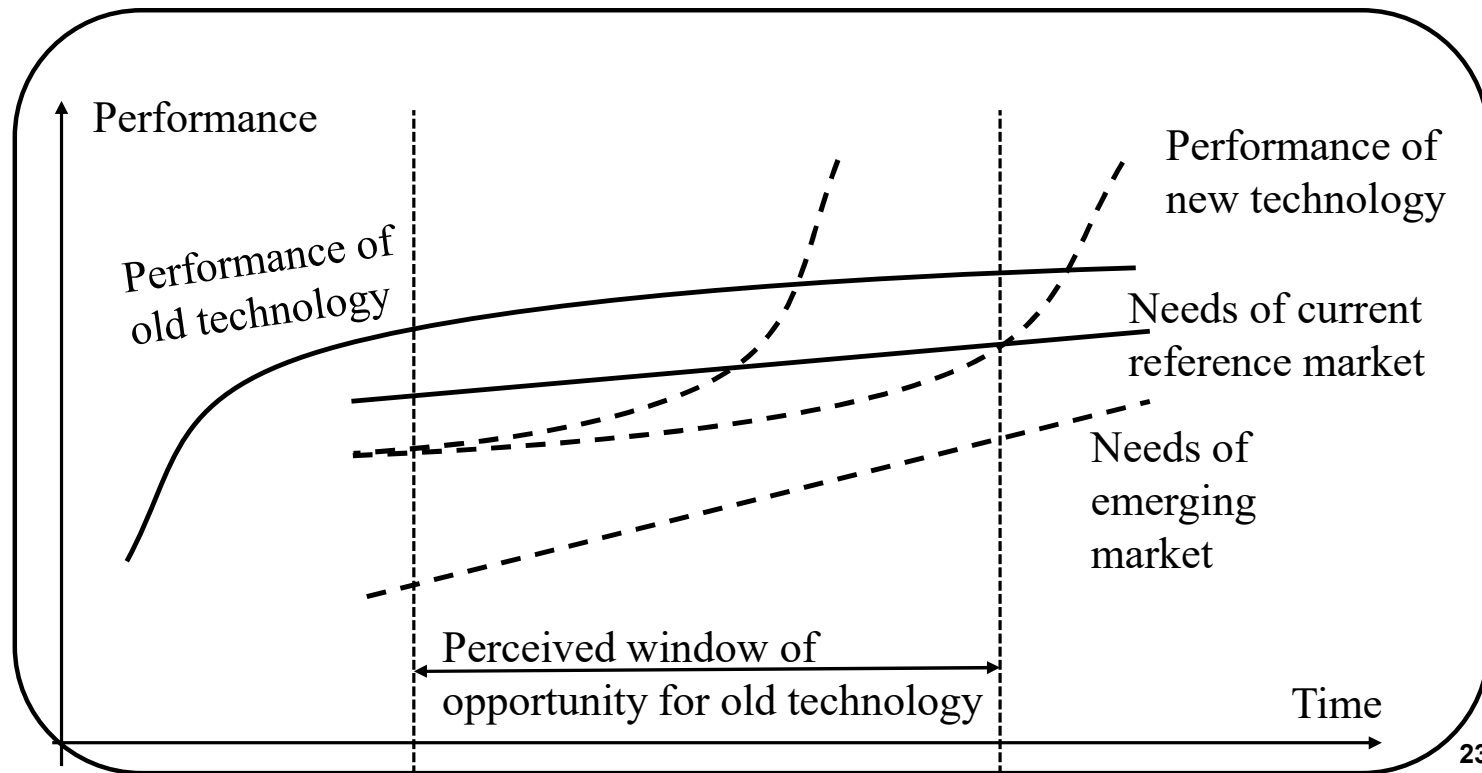


Types of innovation

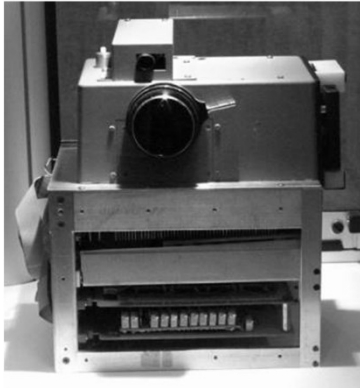
- Radical innovations sometimes are disruptive and sometimes not
- How can innovations be *disruptive* (i.e. why do *incumbent* firms fail to face them)?

2. Focus on reference markets and customers (Christensen).

Types of innovation



23



Kodak's first prototype,
1975, 4 kg, 0.01 MP



Canon
Powershot 600
1996, 0.5 MP



Kodak Prism XLC
previewing system



Kodak DCS 100,
1991 1.3 MP



Kodak DCS 460
1995, 6 MP

Types of innovation

- Radical innovations sometimes are disruptive and sometimes not
- How can innovations be *disruptive* (i.e. why do *incumbent* firms fail to face them)?

2. Focus on reference markets and customers (Christensen).

Where is the problem?

- resource allocation and project selection are driven by observable market demand
- market research prevails over envisioning of the future
- middle managers can test new technology and try championing innovative projects... but can't "change the givens"
- timid attempts to develop "hybrids" that still leverage existing assets
- "don't kill the cash cow but milk it" attitude (e.g., contrast to Netflix in streaming)

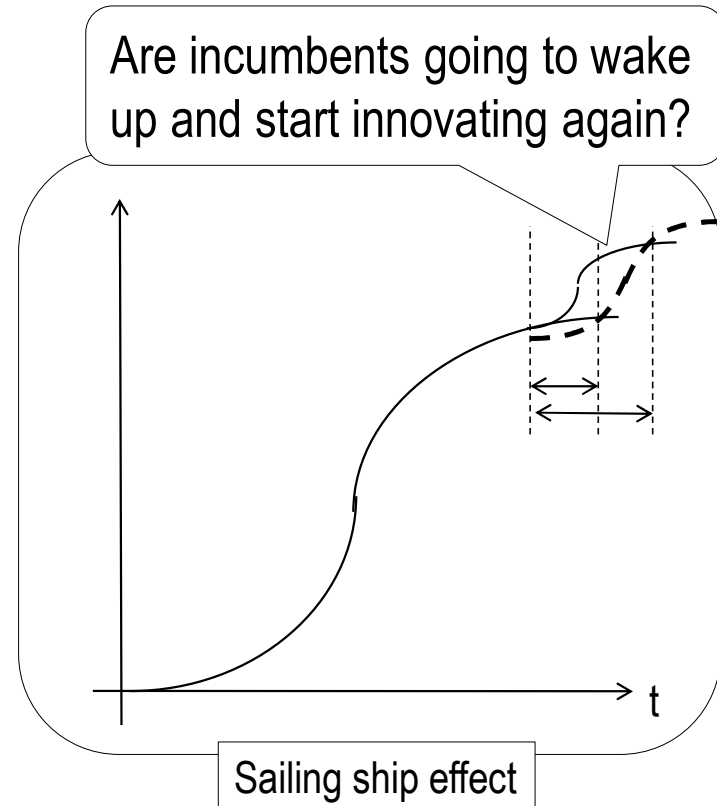
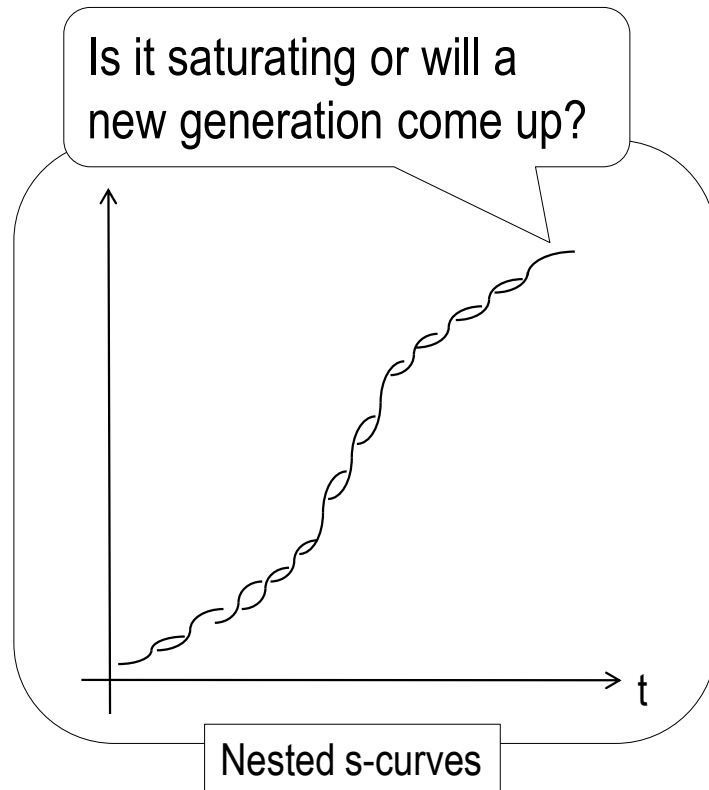


J.M.W. Turner, *The Fighting Temeraire tugged to her last berth to be broken up*, 1838
(National Gallery, London)

Types of innovation

- Radical innovations sometimes are disruptive and sometimes not
- Why can innovations *not become* disruptive (i.e. why do incumbents win over innovating new entrants)?
 1. There are risks in using “maturing” S-curves as forecasting tools

Types of innovation



XIX century innovation in sailing ships

- Retreating to defensible market segments, given pros and cons of the two technologies (coastal? ocean?)
- Improving core technology
 - better charts of winds and currents
 - steel frameworks → larger ships with more payload, more sail, higher critical speed
- Making some use of the new technology
 - machinery allowing less crew (winches powered by “steam donkeys”)
 - steam tugboats



“Lightning” clipper 1854-1869 72 m length



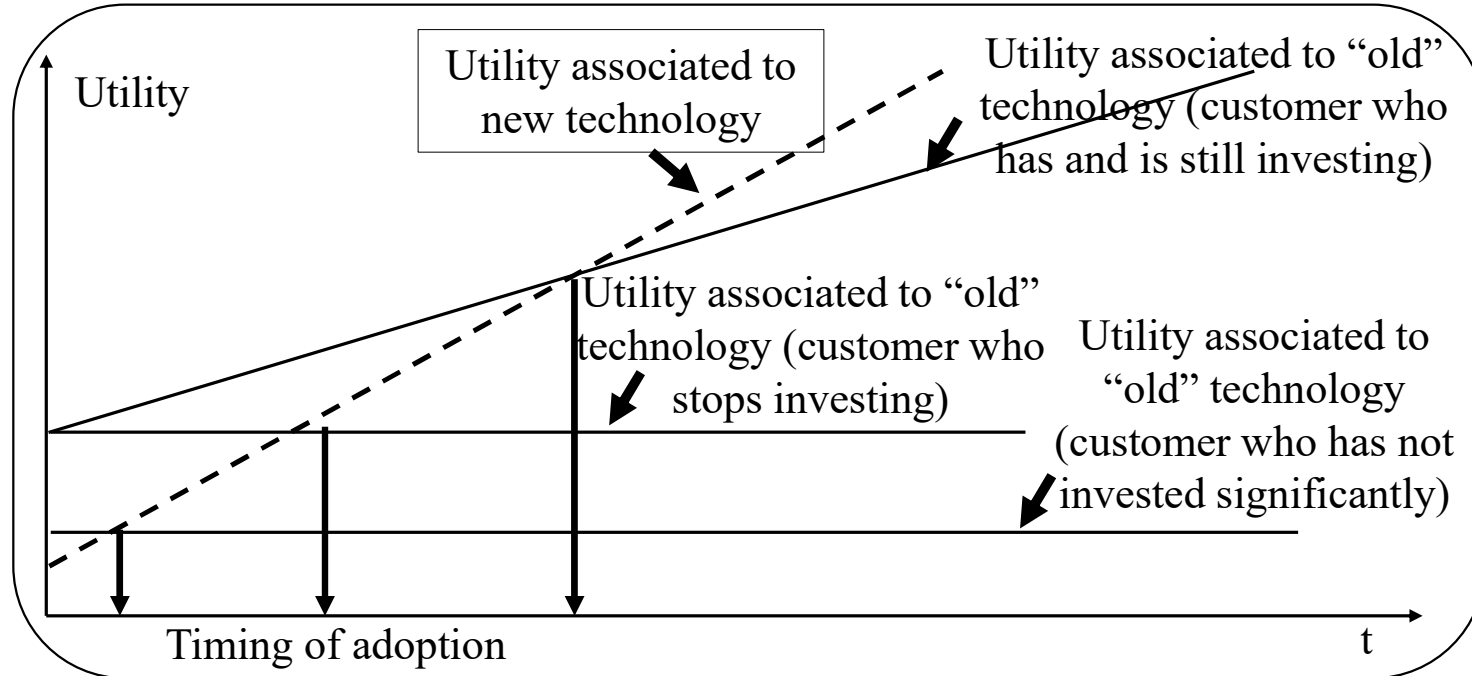
"Preussen" 5 masted windjammer, 1902-1910, 147 m length, 6800 m² sail area

Types of innovation

- Radical innovations sometimes are disruptive and sometimes not
 - Why can innovations *not become* disruptive (i.e. why do incumbents win over innovating new entrants)?
2. The market might reject it
 - a coherent and sustainable paradigm does not emerge
 - the market is locked in the old technology (theory of localized technological change)

Types of innovation

- Radical innovations sometimes are disruptive and sometimes not
- Why can innovations *not become* disruptive (i.e. why do incumbents win over innovating new entrants)?



Types of innovation

- Radical innovations sometimes are disruptive and sometimes not
 - Why can innovations *not become* disruptive (i.e. why do incumbents win over innovating new entrants)?
3. Incumbents might have significant advantages
- Incumbents might successfully imitate entrants
 - Entrants might not commit enough to "paradigm building" and/or be able to secure complementary assets
 - If generic, incumbents might own them
 - If specialized or co-specialized, who is going to build them?



Types of innovation

- Radical innovations sometimes are disruptive and sometimes not
 - Why can innovations *not become* disruptive (i.e. why do incumbents win over innovating new entrants)?
4. It's not really a radical innovation, but a modular one → little impact on firms, supply chains, paradigms
 5. Markets for technology allow incumbents to wait and acquire
 - Role of VC financing
 - Antitrust issues

Sensor diagonal size = 27-43 mm



Sensor diagonal size = 7-15 mm Sensor diagonal size = 5-10 mm

Did you see disruption in
reflex cameras?
What about compact
cameras?
Why?

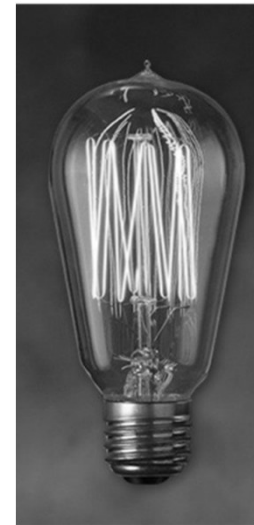


Types of innovation



Suppose you are an incumbent.
What could your strategy be, in order to avoid
being disrupted by radical innovation?

-
-
-
-
-
-
-



Types of innovation



Suppose you are an entrant.
What could your strategy be, in order to
disrupt incumbents?

-
-
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-
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